Zhiyu(Joe) Wang

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WORK EXPERIENCE

Software Development Engineer - AI/ML, Adobe, *Seattle, WA* March 2025 - Present
Firefly Platforms

- **Founding engineer on Adobe Firefly's new inference server**, building a framework-agnostic alternative to an overburdened production system — designing the control plane, Redis-based request queue, and custom scale-to-zero autoscaler, integrating parallel weight download via s5cmd into a purpose-built sidecar, and leveraging filesystem snapshots to minimize cold start latency — enabling researchers to self-serve model deployments across multiple generative models on Kubernetes.
- **Led end-to-end design and delivery of a GPU capacity management system for Adobe Firefly within a team of 4** — spanning a time-series forecasting pipeline for daily GPU demand prediction, an ops control plane, a self-serve UI, and a Grafana dashboard suite for monitoring, reporting, and prediction visualization — standardizing GPU resource management across the org and scaling supported model launches from 5 to 20 simultaneously.
- **Standardized on-call practices for Adobe Firefly's production inference platform** — building runbooks, structured incident note templates, and knowledge-sharing processes to improve response consistency and reduce tribal knowledge dependencies across the team.

Software Development Engineer, AWS, *Seattle, WA* April 2024 - March 2025
Amazon Translate

- Led the build of core Machine translate inference image upgrading. Led the complete process of key dependencies upgrading, including Pytorch, CUDA, etc; Image tests, and production deployment.
- Developed new test scripts for new image accuracy tests and performance load tests;
- Co-Mentored intern to conduct proof of concept project that leverages Amazon Bedrock for Machine translation.

Amazon Bedrock

- Contributed to the launch of the batch inference for Amazon Bedrock, taking ownership of key components and workflows, including batch prompt preprocessing, ensuring efficient and secure operations.
- Led the provisioned throughput metrics dashboard initiation and design. Contributed to the implementation of metrics emitting, aggregating, and presentation of provisioned throughput resources utilization. The clear and accurate data points provided by the project serves as the single sources of truth for GPU resource tracking and cost metering.

Software Development Engineer, Amazon, *Seattle, WA* April 2021 - April 2024

- Led the development of a financial data processing platform for Amazon Stores and Fulfillment Centers Finance.
- Led a crucial performance optimization project, reducing the processing duration from 2 hours to 5 minutes. Contributed to the migration to Serverless OLAP Engine, detailed design and implementation in Java of a workflow orchestrator, and Business Logic modularization in SQL.
- Initiated and led the development of a data exporting feature, offering flexible data granularity and seamless data aggregation, reducing processing time from 3 hours to under 10 minutes. Contributed to the parallelization of the System integration in Java Multi-threading.

Software Engineer, Digi International, *Boston, MA* July 2019 – April 2021

- Contributed to Digi Accelerated Linux Operating System for transportation routers, including router OS network features and WebUI.

EDUCATION

Boston University, Boston, MA **GPA: 3.60** Sep 2017 - May 2019
Master of Science in Electrical and Computer Engineering, *Graduate with Intern Practice.*

SKILLS

Programming Languages: Python, Java, Rust, TypeScript, SQL, Bash.
Databases: PostgreSQL, Radis, AWS DynamoDB**,** AWS redshift
Technologies/Frameworks:Kubernetes, Ollama, React, MUI, S3, AWSLambda, Amazon Quicksight, Pytorch, Glue, Athena, CloudWatch, Docker, boto3, JupyterNotebook, Git.